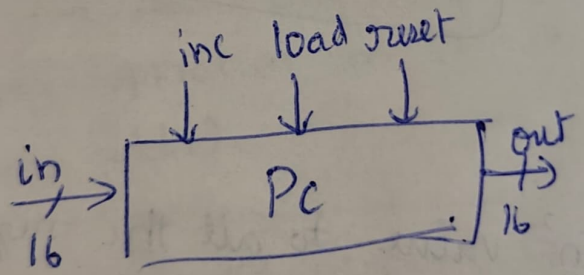
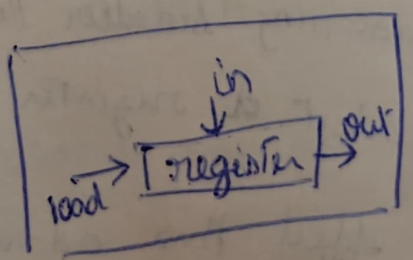
Program Counter:

* It is a priority logic.

if reset(t): out[t+1] = 0
else if load(t): out[t+1] = in(t)
else if inc(t): out[t+1] = out[t] + 1
else out[t+1] = out[t]



Logically
→ reset has highest priority
because when reset == 0, irrespective of 'load' and 'inc', my Register inside PC must be 0



↳ * needed to store the value and perform an operation (reset/inc/load)

* Something has to remember present state

If reset == 1 , out = 0 — Priority: reset ↑ >> load >> inc

else

load = 1 , out = in

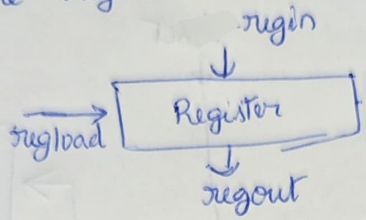
else

inc = 1 , out = present state + 1

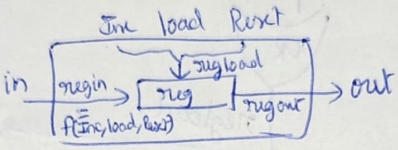
else

out = present state

* Consider the register in the PC



whatever changes I make to my present state, I must save it in this register.

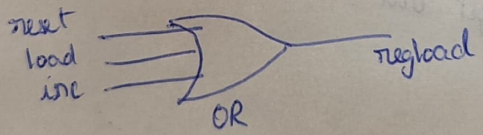


and if I am not performing any operation, the present state must be outputted as it is

⇒ reset = 0 & inc = 0 & load = 0.

If any one is 1 → O/P gets modified.

↳ can be realised using OR gate.



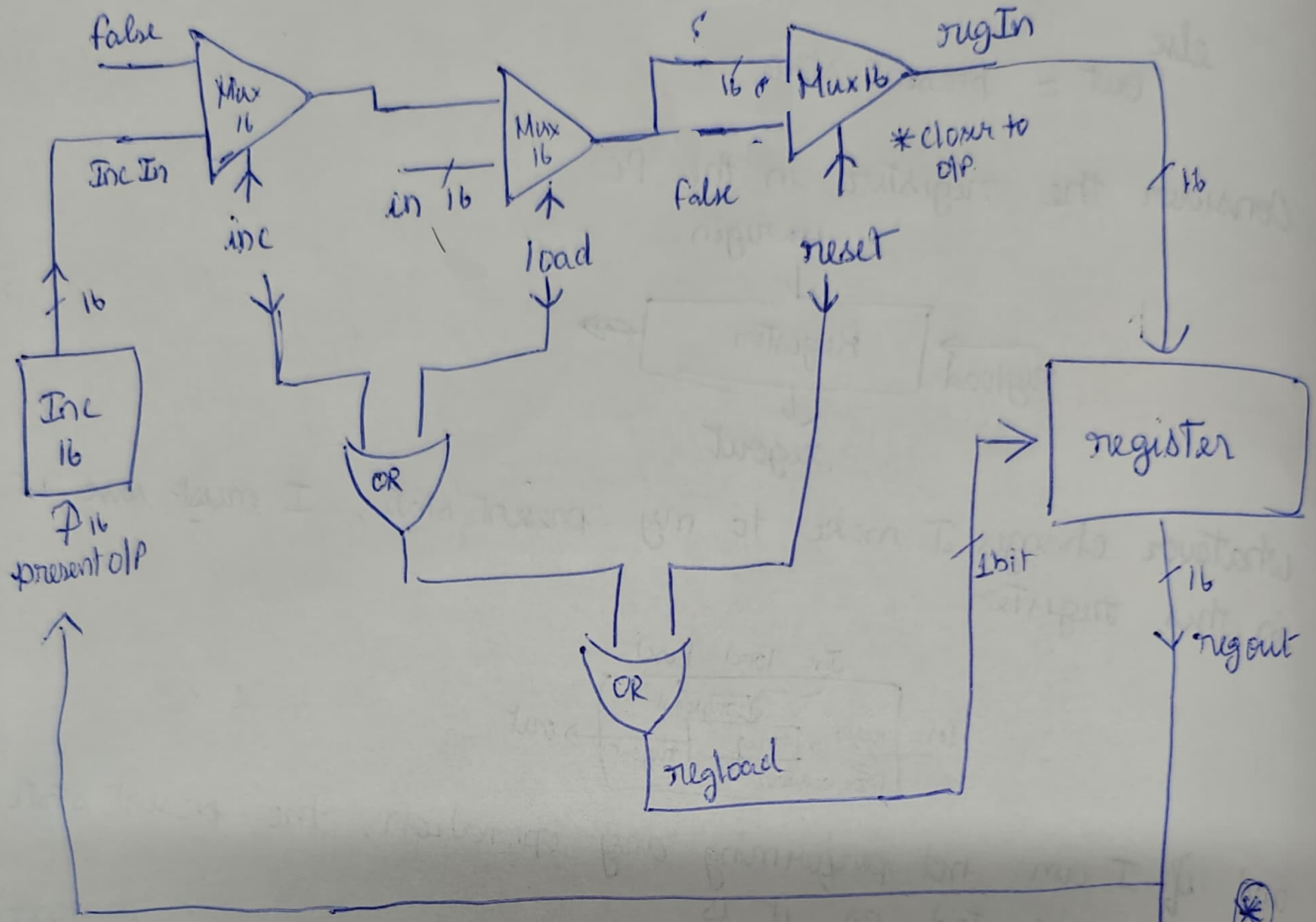
According to priority, reset >> load >> inc

TT

Inc	load	Reset	Output
0	0	0	present State
X	X	1	0000 / all 0's
X	1	0	load in [i]
1	0	0	present state + 1

* when designing → highest priority → closer to O/P

Full Design of a Program Counter



In this HDL, out port cannot be directly used as an input

* So we are making regout as intermediate port and used 2 Inv to get out.